

What factors most strongly drive burger restaurant ratings in Philadelphia



By Oliver Marszalek and Alan Kim

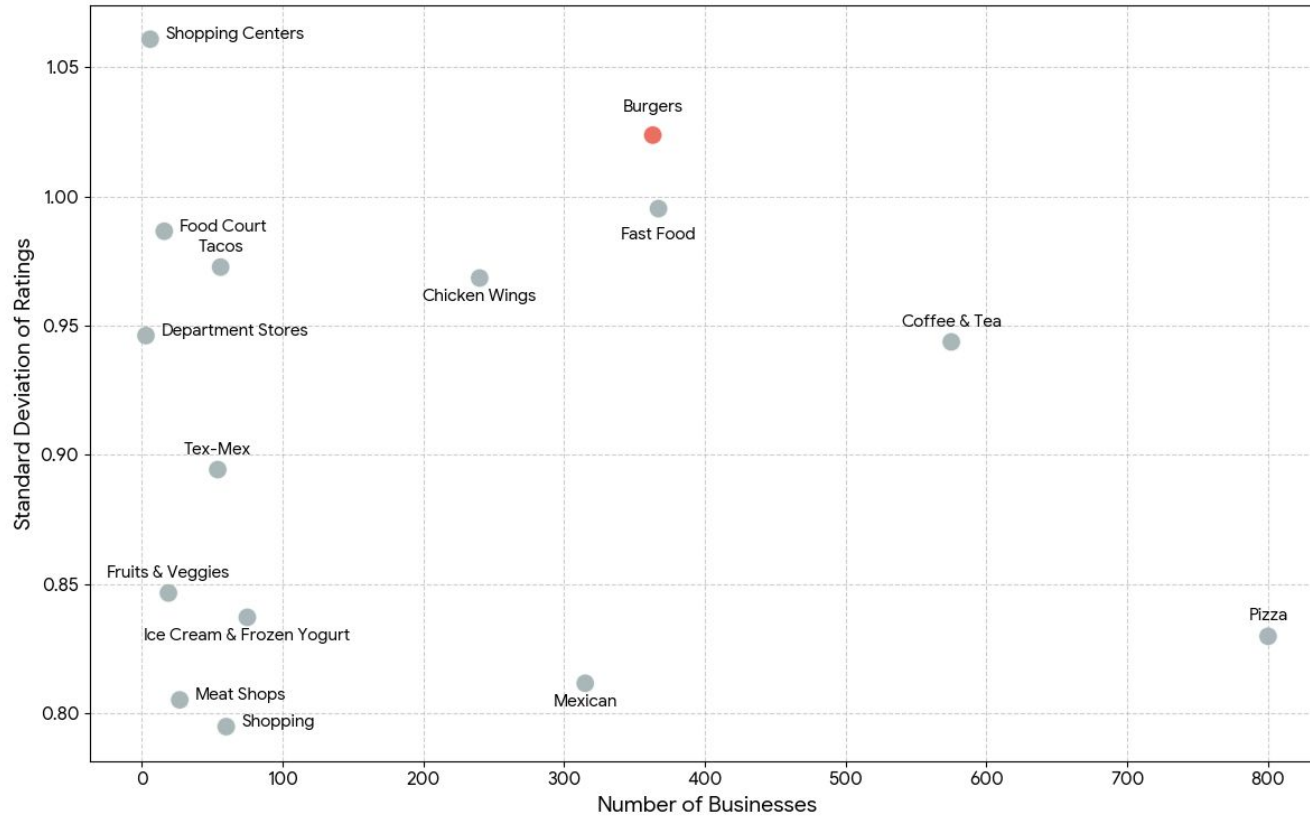


Burger restaurants vary in quality

Top Categories by Standard Deviation

Category	# of Business	# of Reviews	Average Rating	Standard Deviation of Rating
Shopping Centers	6	6128	4.518	1.060
Burgers	363	41746	3.568	1.023
Fast Food	367	26889	3.501	0.995
Food Court	16	6428	4.555	0.986
Tacos	56	5951	3.852	0.972
Chicken Wings	240	14869	3.461	0.968
Department Stores	3	5965	4.549	0.946
Coffee & Tea	575	67688	4.002	0.943
Tex-Mex	54	8717	3.449	0.894
Fruits & Veggies	19	7991	4.405	0.846
Ice Cream & Frozen Yogurt	75	14121	4.129	0.837
Pizza	800	68523	3.663	0.829
Mexican	315	42210	3.792	0.811
Meat Shops	27	7597	4.465	0.805

Market Opportunity (Top 15 Categories)



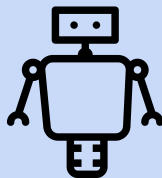
68% of Burger Places in Philadelphia

2.54 (Poor / Mediocre) - 4.59 (Excellent)



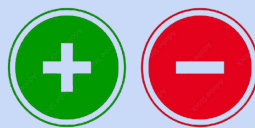


Exploring user data with machine learning



Creating baseline categories

Create key words → Assign each key word into categories → divide categories into positive and negative.



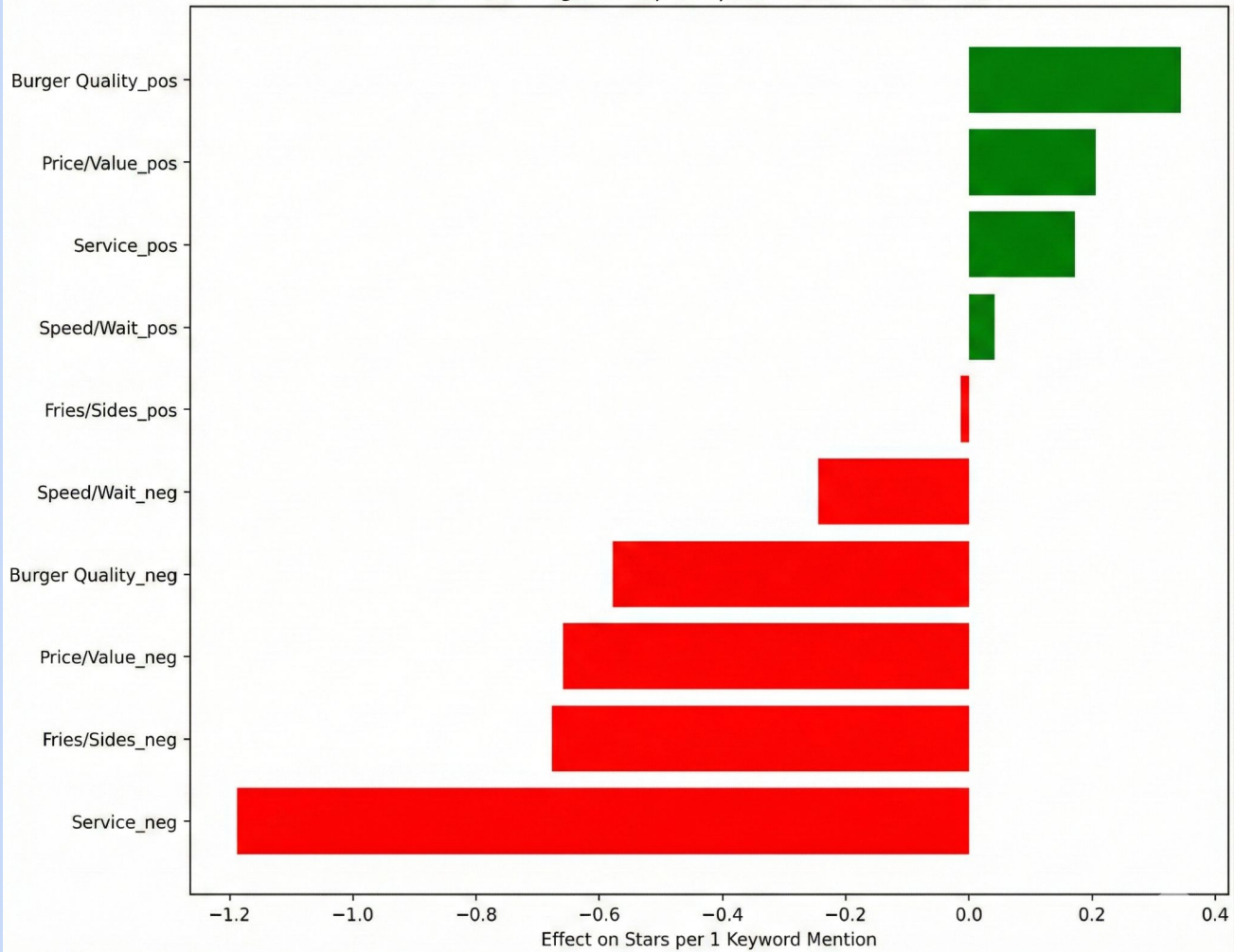
Extracting text and make meaningful connections

Extract reviews as text → Match each review to stars → Filter out key words → Have the ML model analyze the weight of each keyword



Regression model: $\hat{y} = b + w_1x_1 + w_2x_2 + \dots + w_nx_n$

Positive vs Negative Topic Impact on Stars (Coefficients)



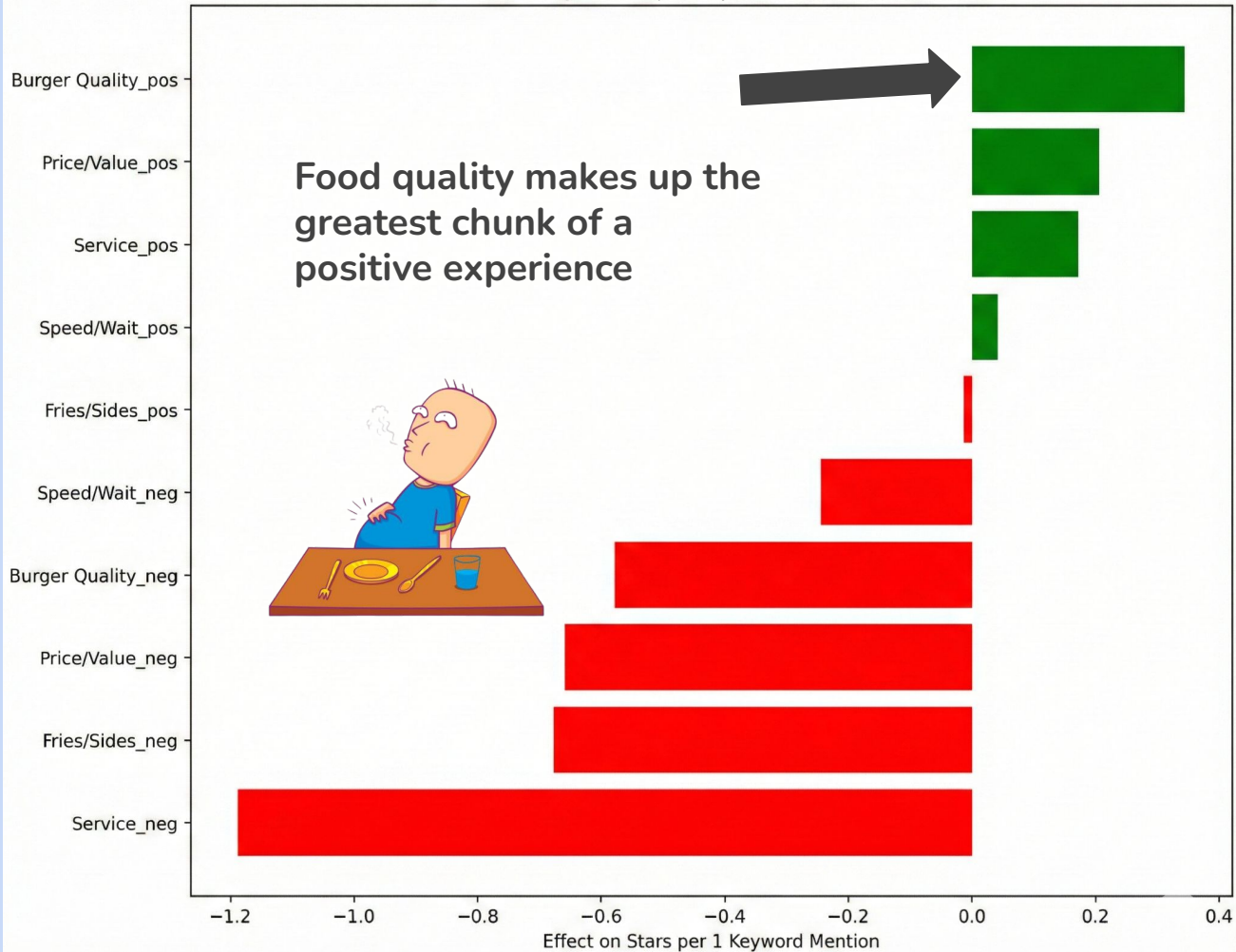
Positive vs Negative Topic Impact on Stars (Coefficients)



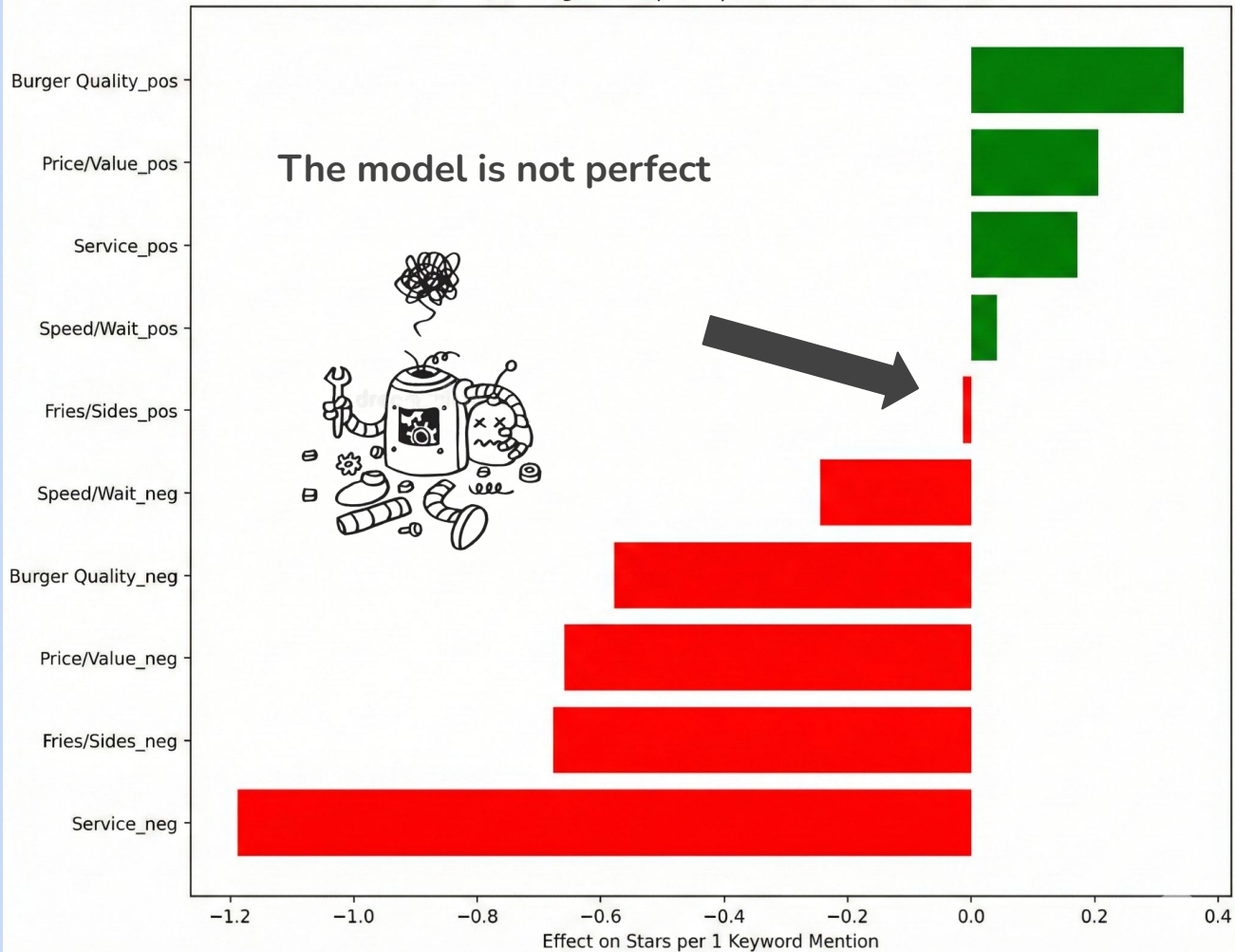
Positive vs Negative Topic Impact on Stars (Coefficients)



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Positive vs Negative Topic Impact on Stars (Coefficients)





Pitfalls along the way

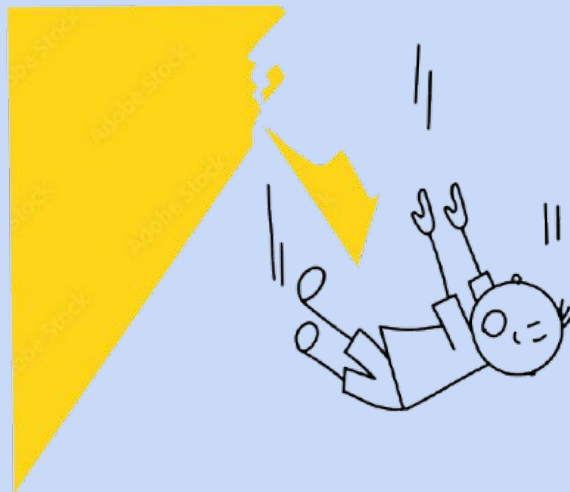
Initial model

- Attributes instead of user reviews
- Focused too much on location
- Included unwanted data

Result: Terrible R^2 (~ 0)
Terrible MAE ($\sim .9$)

Current

Result: Improved R^2 ($\sim .68$)
Improved MAE ($\sim .63$)





Best Locations For Burger Restaurants

Neighborhoods with Best Values

Neighborhood	Average Rating	Review Counts	Average Rent per Sqft	Score
Southwest Philly	4.077	13	18.0	9.234
West Philly	3.91	378	18.0	8.494
North Philly	3.102	147	15.0	6.415
Northeast Philly	3.838	722	23.0	6.404
Holmesburg	3.388	206	18.0	6.378
East Falls	3.87	524	25.0	5.991
Overbrook	3.427	454	20.0	5.873
West Oak Lane	3.125	24	18.0	5.425
Manayunk	4.159	906	32.5	5.322
South Philly	3.917	1585	30.0	5.115

$$Score = \frac{(Average\ Rating)^2}{Average\ Rent}$$

Counter Arguments



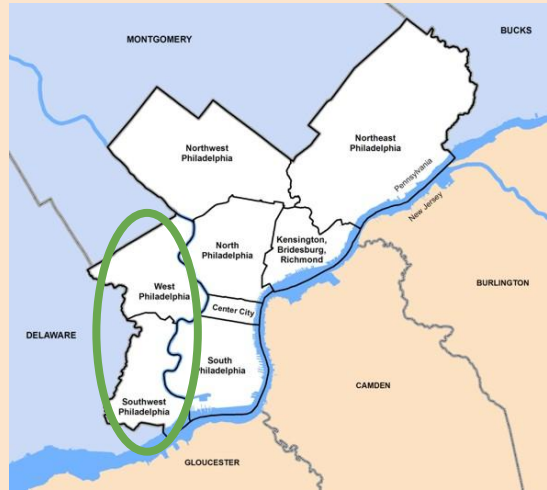
1. Regression Model shows only correlation
2. Limiting reviews based off of selected keywords
3. Reviews are bias and wording is subjective

Conclusion and our recommendations

Clean up **negative** categories



Move and/or expand here:



Hire **excellent** burger makers





Thank You!